STRUCTURAL INVESTIGATION OF STRATUM CORNEUM MODEL MEMBRANES BY NEUTRON DIFFRACTION

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ABSTRACT

In the "brick-and-mortar" structure of the stratum corneum, corneocytes form the bricks while the intercellular lipid matrix, composed primarily of ceramides, cholesterol, and free fatty acids, acts as the vital mortar protecting against trans-epidermal water loss and other external hazards[1]. Elucidating the detailed organization of these intralamellar lipid layers is an essential prerequisite when investigating how the surrounding lipids interact structurally with the corneocyte lipid envelope. In particular, understanding the coexistence of the short periodicity phase (SPP) and long periodicity phase (LPP) alongside the presence of cholesterol crystals is critical for evaluating skin barrier properties[2]. In this work, we present the neutron diffraction results on model lipid membranes combining cholesterol, behenic acid, and ceramides. Specifically, we selected ceramides: CER[NP], CER[NS], and a 50/50 mixture of both. Our diffraction profiles reveal how modifying ceramide composition impacts lamellar repeat distances, phase separation, and the onset of crystalline cholesterol domains, providing a solid structural foundation for advanced skin barrier and transdermal delivery models.

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